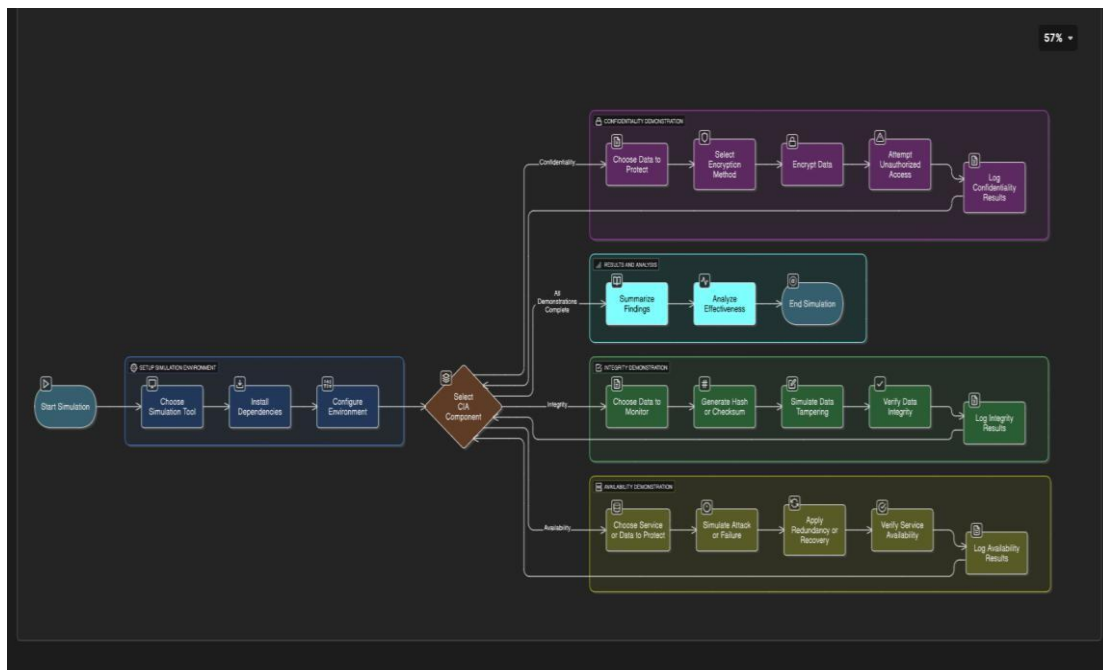




DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

DEPARTMENT	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING				
Semester: VII	Course Title: Data Security & Privacy Lab			Course Code: 22AI73	
PROGRAM QUESTION	1) Exploring CIA Triad with Simulations: using a Python-based simulation or online tools to demonstrate how data can be protected (confidentiality), checked for tampering (integrity), and ensured to be available under attack (availability).				
STUDENT NAMES	Udayaraj		USN	1DS22AI057	
LAB MARKS	Program Execution (10)	Source Code (10)	Result (5)	Innovation/ Extra Features (5)	Total
TOOLS USED	Cursor				
MAP COURSE OUTCOME	CO1				
MAP PROGRAM OUTCOME	PO2, PO3, PO4, PO5				
MAP PROGRAM SPECIFIC OUTCOME	PSO1, PSO2				
UML / SEQUENCE DIGRAM					
COURSE COORDINATOR	Dr ARUNA M G & Prof ENSTEIH SILVIA				

Code for Program 1:

1) Exploring CIA Triad with Simulations: using a Python-based simulation or online tools to demonstrate how data can be protected (confidentiality), checked for tampering (integrity), and ensured to be available under attack (availability).

```
import streamlit as st, hashlib, secrets, time, random, threading
from cryptography.fernet import Fernet

st.set_page_config(page_title="CIA Triad Demo", page_icon="🔒", layout="wide")

class CIADemo:
    def __init__(self):
        if "cipher" not in st.session_state:
            key = Fernet.generate_key()
            st.session_state.cipher = Fernet(key)
            st.session_state.encrypted, st.session_state.hash = b"", ""
            st.session_state.availability = True
            st.session_state.server_health = {s: {"status": "online", "load": l}
                                                for s, l in zip(["Primary", "Backup1", "Backup2"], [30, 15, 10])}

# --- Confidentiality ---
def confidentiality(self):
    st.header("🔒 Confidentiality")
    data = st.text_area("Sensitive data:", "Credit card: 1234-5678-9012-3456")
    c1, c2 = st.columns(2)
    with c1:
        if st.button("Encrypt"):
            st.session_state.encrypted = st.session_state.cipher.encrypt(data.encode())
            st.code(st.session_state.encrypted, language="text")
    with c2:
        if st.session_state.encrypted:
            if st.button("Decrypt (correct key)"):
                st.success(st.session_state.cipher.decrypt(st.session_state.encrypted).decode())
            if st.button("Try Wrong Key"):
                try: Fernet(Fernet.generate_key()).decrypt(st.session_state.encrypted)
                except: st.success("Wrong key rejected")

# --- Integrity ---
def integrity(self):
    st.header("🔍 Integrity")
    data = st.text_area("Enter data:", "Important data")
    if st.button("Generate Hash"):
        st.session_state.hash = hashlib.sha256(data.encode()).hexdigest()
        st.session_state.data_ref = data
        st.code(st.session_state.hash, language="text")
```

```

if st.session_state.hash:
    check = st.text_area("Verify data:", st.session_state.data_ref)
    if st.button("Verify"):
        ok = hashlib.sha256(check.encode()).hexdigest()==st.session_state.hash
        st.success("✅ Integrity OK") if ok else st.error("⚠️ Tampered!")

```

--- Availability ---

```
def availability(self):
```

```
    st.header("🔗 Availability")
```

```
    c1,c2=st.columns(2)
```

```
    with c1:
```

```
        st.metric("Uptime", "99.95%" if st.session_state.availability else "85%")
```

```
        for s,stats in st.session_state.server_health.items():
```

```
            st.write(f'{"✅" if stats["status"]=="online" else "⚠️"} {s} Load: {stats["load"]} %')
```

```
def main():
```

```
    demo=CIADemo()
```

```
    st.sidebar.title("Navigation")
```

```
    choice=st.sidebar.radio("Section",["Overview","Confidentiality","Integrity","Availability","Password Security"])
```

```
    if choice=="Overview":
```

```
        st.title("CIA Triad"); st.write("Confidentiality 🔒 | Integrity 🔑 | Availability 🔗")
```

```
    elif choice=="Confidentiality": demo.confidentiality()
```

```
    elif choice=="Integrity": demo.integrity()
```

```
    elif choice=="Availability": demo.availability()
```

```
    elif choice=="Password Security": demo.password()
```

```
    st.sidebar.markdown("---"); st.sidebar.info("Educational demo. Not for production use.")
```

RESULT:

